Extra Bite #7: Software Defined Radio and DSP Basics

Direct sampling, Nyquist, decimation, FFT, FIR filters. The vocabulary of a software radio.

Group E7F · 14 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

What An SDR Actually Does

A software-defined radio replaces most of the traditional analog signal chain with one fast **analog-to-digital converter (ADC)** followed by software. Once the RF is digitized, every filter, mixer, demodulator, and equalizer is just math. E7F asks you to recognize the vocabulary of that math.

Direct Sampling And The Nyquist Rule

Direct sampling means the incoming RF is digitized by an ADC **without first being mixed down to a lower frequency by a local oscillator**. (E7F01) No analog mixer; the ADC just slams the RF straight into bits.

Nyquist rule: to reproduce an analog signal accurately, sample it at **at least twice the rate of the highest frequency component**. (E7F05) Anything lower and high-frequency content "folds" back into the digitized signal as false low-frequency content (aliasing).

Two consequences for a direct-sampling SDR:

- **Sample rate** sets the maximum receive bandwidth. (E7F10)
- **Reference voltage level and sample width in bits** set the minimum detectable signal in the absence of noise. (E7F11) More bits = finer voltage steps = quieter noise floor.

Bit Depth Worked Example (E7F06)

How many bits are needed to sample a 1 V signal with 1 mV resolution?
1 volt / 1 millivolt = **1,000 steps**. Number of bits N must satisfy $2^N \geq 1{,}000$. $2^{10} = 1{,}024 \geq 1{,}000$, so **10 bits** is the minimum.

DSP Filter Vocabulary

Term	What it does	Q
Adaptive filter	A digital filter that learns and removes **unwanted noise from a received SSB signal** automatically. The classic SDR noise-reduction tool.	E7F02
Hilbert-transform filter	A DSP filter used to **generate an SSB signal** by producing the 90°-shifted (quadrature) copy needed for the phasing method.	E7F03
SSB generation by DSP	Signals are **combined in quadrature phase relationship** – the in-phase (I) and the 90°-shifted (Q) audio are added, which cancels one sideband and keeps the other.	E7F04
FFT (Fast Fourier Transform)	Converts signals from the **time domain to the frequency domain** – turns a stream of samples into a spectrum display.	E7F07
FIR (Finite Impulse Response) filter	A standard DSP filter that can **delay all frequency components of the signal by the same amount** (linear phase). Very useful for digital modes where group delay matters.	E7F12
Taps	Stages in the FIR filter that **provide incremental signal delays for the filter algorithm**. Each tap multiplies a delayed sample by a coefficient and sums. **More taps = sharper filter response** (E7F14).	E7F13, E7F14

Decimation

Term	What it does	Q
Decimation	**Reduces the effective sample rate by removing samples.** (E7F08) Used to cut the data rate after the ADC once the signal of interest is selected by a digital filter.	E7F08
Anti-aliasing filter (before decimation)	Removes high-frequency signal components that would otherwise be reproduced as **false lower-frequency components** after the rate is dropped. (E7F09)	E7F09

Mental Cheat Sheet

- Direct sampling = digitize the RF without a mixer/LO step.
- Nyquist: sample at **at least 2x** the highest frequency.
- Sample rate sets the SDR receive bandwidth.
- Bits + reference voltage set the minimum detectable signal.
- To resolve 1 mV in 1 V you need 10 bits ($2^{10} = 1024 > 1000$).
- Adaptive filter → removes SSB noise.
- Hilbert-transform filter → generates SSB.
- SSB by DSP = combine signals in **quadrature**.
- FFT = time domain → frequency domain.
- FIR filter strength: linear phase (equal delay for all components).
- Taps = delay stages in a DSP filter. More taps = sharper filter.
- Decimation = remove samples to lower the sample rate.
- Anti-aliasing filter before decimation kills the high frequencies that would alias back.

Common Traps

- Direct sampling means **no** analog mixer. The trap answer talks about switching mixers or about generating I/Q without RF amplification.
- Nyquist is **2x**, not 1x or “the same rate.” The trap answer says “at the same rate” or “at least half the rate.”
- Decimation **removes** samples to drop the rate; it does not “attenuate the signal” or “convert to binary-coded decimal.”
- FFT converts **time to frequency**, not analog to digital (that is the ADC’s job).
- SSB via DSP uses **quadrature combination**, not mixing products converted to voltages.
- Anti-aliasing filter is needed because aliasing creates **false low-frequency components**, not because of any DC restoration issue.

Bite 7 Question Bank (E7F)

E7F01. What is meant by “direct sampling” in software defined radios?
A. Software is converted from source code to object code during operation of the receiver.
B. I and Q signals are generated by digital processing without the use of RF amplification.
C. Incoming RF is digitized by an analog-to-digital converter without being mixed with a local oscillator signal.
D. A switching mixer is used to generate I and Q signals directly from the RF input.

E7F02. What kind of digital signal processing audio filter is used to remove unwanted noise from a received SSB signal?
A. An adaptive filter.
B. A crystal-lattice filter.
C. A Hilbert-transform filter.
D. A phase-inverting filter.

E7F03. What type of digital signal processing filter is used to generate an SSB signal?
A. An adaptive filter.
B. A notch filter.
C. A Hilbert-transform filter.
D. An elliptical filter.

E7F04. Which method generates an SSB signal using digital signal processing?
A. Mixing products are converted to voltages and subtracted by adder circuits.
B. A frequency synthesizer removes unwanted sidebands.
C. Varying quartz crystal characteristics are emulated in digital form.
D. Signals are combined in quadrature phase relationship.

E7F05. How frequently must an analog signal be sampled to be accurately reproduced?
A. At least half the rate of the highest frequency component of the signal.
B. At least twice the rate of the highest frequency component of the signal.
C. At the same rate as the highest frequency component of the signal.
D. At four times the rate of the highest frequency component of the signal.

E7F06. What is the minimum number of bits required to sample a signal with a range of 1 volt at a resolution of 1 millivolt?
A. 4 bits.
B. 6 bits.
C. 8 bits.
D. 10 bits.

E7F07. What function is performed by a Fast Fourier Transform?
A. Converting analog signals to digital form.
B. Converting digital signals to analog form.
C. Converting signals from the time domain to the frequency domain.
D. Converting signals from the frequency domain to the time domain.

E7F08. What is the function of decimation?
A. Converting data to binary-coded decimal form.
B. Reducing the effective sample rate by removing samples.
C. Attenuating the signal.
D. Removing unnecessary significant digits.

E7F09. Why is an anti-aliasing filter required in a decimator?
A. It removes high-frequency signal components that would otherwise be reproduced as lower frequency components.
B. It peaks the response of the decimator, improving bandwidth.
C. It removes low-frequency signal components to eliminate the need for DC restoration.
D. It notches out the sampling frequency to avoid sampling errors.

E7F10. What aspect of receiver analog-to-digital conversion determines the maximum receive bandwidth of a direct-sampling software defined radio (SDR)?
A. Sample rate.
B. Sample width in bits.
C. Integral non-linearity.
D. Differential non-linearity.

E7F11. What sets the minimum detectable signal level for a direct-sampling software defined receiver in the absence of atmospheric or thermal noise?
A. Sample clock phase noise.
B. Reference voltage level and sample width in bits.
C. Data storage transfer rate.
D. Missing codes and jitter.

E7F12. Which of the following is generally true of Finite Impulse Response (FIR) filters?
A. FIR filters can delay all frequency components of the signal by the same amount.
B. FIR filters are easier to implement for a given set of passband rolloff requirements.
C. FIR filters can respond faster to impulses.
D. All these choices are correct.

E7F13. What is the function of taps in a digital signal processing filter?
A. To reduce excess signal pressure levels.
B. Provide access for debugging software.
C. Select the point at which baseband signals are generated.
D. Provide incremental signal delays for filter algorithms.

E7F14. Which of the following would allow a digital signal processing filter to create a sharper filter response?
A. Higher data rate.
B. More taps.
C. Lower Q.
D. Double-precision math routines.

Answer Key

E7F01: C	E7F02: A	E7F03: C	E7F04: D
E7F05: B	E7F06: D	E7F07: C	E7F08: B
E7F09: A	E7F10: A	E7F11: B	E7F12: A
E7F13: D	E7F14: B		